When does the Past Hypothesis select the subsystems? Quantum mereology as constrained tomography

Claude (Anthropic)
produced in an extended conversation with a human user, who directed the project

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Abstract

In Hilbert-space fundamentalism the world is a Hamiltonian H and a state ψ, and the division into subsystems must be derived. Stoica has shown that no such derivation can be unique: every tensor product structure (TPS) in which H is local is one of a continuum related by unitaries commuting with H. We show that imposing a strengthened form of the Past Hypothesis as a selection rule — choose the division in which some moment of the history is simple (close to a product state, or to a matrix product state) — reduces the preferred-structure problem to constrained quantum state tomography: recovering the simple moment of the history (the *bounce*) from a single projective measurement in the energy eigenbasis, with simplicity as prior information. Known results on constrained tomography then imply that, for a generic eigenbasis and an exactly simple bounce, the rule selects the actual history uniquely up to time translation, a residual freedom whose significance is itself debated. For Haar-random eigenbases we prove a robust version with explicit constants, covering bounces close to matrix product states and hence area-law states with extensive entanglement, for any spectrum, including the commensurable spectra recently proposed to avoid Boltzmann brains. For local Hamiltonians we show that central-limit-level information about energy distributions cannot separate simple states that differ locally, and that under an eigenstate-thermalization-type hypothesis simple states are separated in expectation. The uniform separation the rule needs is open; simulations of chaotic spin chains of up to 12 qubits support it at high energy. The selection is thus proved for generic and random eigenbases and conjectural for local Hamiltonians. For time-reversal-invariant Hamiltonians we prove the analogous exact statement with time reversal added to the residual freedom: the rule fixes the history but not its direction.

Status and provenance. Not peer reviewed. The results, proofs, code and text were produced by an AI system, Claude (Anthropic), in an extended conversation with a human user, who chose the direction and requested checks but did not derive results. No human expert has checked the proofs. Novelty is unverified: during this work the model twice claimed novelty that literature searches then refuted (Stoica's Remark 11; the Kech–Wolf theorem), and the use of the Past Hypothesis as a selection principle was first proposed by Stoica (Sec. VII.B of his 2022 paper). Appendix B summarizes the checks performed; the extended version records every check and every error found. Code and raw data accompany the manuscript.

1 Introduction

Hilbert-space fundamentalism (HSF) holds that the only fundamental data are a Hilbert space, a Hamiltonian H and a state ψ, and that everything else, including the division into subsystems, emerges from them [5, 6]. That subsystems are relative to a choice of observables goes back to

Zanardi, Lidar and Lloyd [11]. The factorization problem has since been approached through locality [14], quasi-classical dynamics [5], minimal scrambling [12], classical subsystems [13], invariants of (H, ψ) [10] and spectral data [32, 33, 34]. Stoica has argued that any structure built from (H, ψ) alone in a unitarily covariant way is non-unique: the torus Γ of unitaries commuting with H maps one local TPS to a continuum of equally local ones, in each of which the state reads differently [6, 7, 8]. Soulas, Franzmann and Di Biagio reply that invariant properties of (H, ψ) can fix a TPS, while conceding that their construction lacks direct physical meaning [10]; Stoica maintains that this fixes the structure by hand [9].

Two observations frame this note. For generic spectra the time-evolution orbit is dense in Γ, so alternative local TPSs are limits of time translates: other times rather than other worlds (Stoica [6, Remark 11], credited to D. Ranard). But the finite-dimensional cyclic cosmology of Carroll, Diachenko and Dulani uses a commensurable spectrum, which closes the history into a loop, precisely to avoid Boltzmann brains [15]. The loop has measure zero in Γ, so almost every alternative local TPS reads the state as one the history never visits, and the "which world" problem returns.

Contributions. We study a selection rule based on a strengthened form of the Past Hypothesis: the requirement that the history contains a *simple* moment, which we call the *bounce*. This is a substantive postulate about the early state, close to the Entanglement Past Hypothesis [3] and stronger than low thermodynamic entropy. Stoica already considered the Past Hypothesis for this purpose and argued that it cannot give uniqueness [6, Sec. VII.B]. We show that, among locality-preserving divisions, it gives uniqueness up to time translation (and, for time-reversal-invariant Hamiltonians, time reversal), the case Stoica identifies as yielding a unique spacetime [6, Remark 10 and Sec. VII.E].

1. *Reduction (Sec. 3).* The readings of ψ across all equally local TPSs are exactly the states sharing its energy-weight profile. Selecting a TPS by simplicity is therefore constrained tomography of the bounce from one measurement in the energy eigenbasis. The residual freedom is always at least time translation.

2. *Exact selection (Thm. 5).* By results of Heinosaari, Mazzarella and Wolf [16] and Kech and Wolf [17], a generic eigenbasis identifies an exactly simple bounce once $N - 1$ exceeds twice the dimension of the simple family: from 5 qubits for product states and from 7 for bond-dimension-2 matrix product states (MPS). The rule then selects exactly the history's time translates. For generic real eigenbases, the time-reversal-invariant case, the same thresholds give selection up to time translation and time reversal (Thm. 6).

3. *Robust selection for random eigenbases (Thms. 9 and 10).* For a Haar-random eigenbasis and any spectrum, with probability $1 - e^{-\Omega(2^n)}$ every division at least as simple as the true one reproduces the whole history, up to a time shift, within trace distance $r_0 + 2\eta$, where η^2 is the bounce's distance-squared to the simple family. The MPS version covers area-law bounces with extensive entanglement.

4. *Local Hamiltonians (Sec. 5).* Central-limit information cannot separate product states that differ locally (Prop. 12), so any proof needs eigenvector fine structure. A random-amplitude hypothesis of eigenstate-thermalization (ETH) type gives pairwise separation in expectation (Prop. 14); the uniform separation the rule needs remains open. Its predictions match exact values on chaotic chains of 8–11 qubits to within about 3%.

The mathematical core of item 2 is known, and the technique in item 3 is standard in phase retrieval [18, 19]. What we believe is new is the reduction, the robustness statements including area-law bounces, the analysis of local Hamiltonians, and the consequences for the preferred-structure debate and for commensurable-spectrum cosmologies.

2 Setup

Let $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, $N = 2^n$, and let $H = \sum_k E_k |k\rangle\langle k|$ have non-degenerate spectrum. Let T_0 be a TPS in which H is K-local. For a state ψ define its *energy profile* $W(\psi) \in \Delta^{N-1}$ by $W(\psi)_k = |\langle k|\psi\rangle|^2$, the outcome distribution of an energy measurement. We use the trace distance between pure states, $D(\psi, \phi) = \sqrt{1 - |\langle\psi|\phi\rangle|^2}$, and the Hellinger distance between distributions, $\mathrm{Hel}(p, p')^2 = 1 - \sum_k \sqrt{p_k p'_k} = \frac{1}{2}\|\sqrt{p} - \sqrt{p'}\|_2^2$, which is a metric.

Let $\Gamma = \{V_\varphi = \sum_k e^{i\varphi_k}|k\rangle\langle k|\}$ be the torus of unitaries commuting with H. For every $V \in \Gamma$, H is exactly as local in $T_V := VT_0V^\dagger$ as in T_0, since $V^\dagger H V = H$. Under the genericity assumptions of Cotler, Penington and Ranard [14] these are all the K-local TPSs [10]; our results concern the family $\{T_V\}$ in any case. Expressed in T_0-coordinates, the state ψ viewed through T_V is the *reading* $V^\dagger\psi$. Time evolution is $U(t) = V_{\varphi=-Et} \in \Gamma$, so time translates of T_0 belong to the family.

Lemma 1 (Readings are the profile fibre)**.** $\{V^\dagger\psi : V \in \Gamma\} = \{\chi : W(\chi) = W(\psi)\}$*, up to global phases.*

Proof. $|\langle k|V^\dagger\psi\rangle| = |\langle k|\psi\rangle|$ for all k. Conversely, if $W(\chi) = W(\psi)$, choose φ_k with $\langle k|\chi\rangle = e^{-i\varphi_k}\langle k|\psi\rangle$ wherever these are non-zero, and arbitrarily elsewhere. □

Simplicity. A *simple family* $\mathcal{S}(T)$ is defined relative to a TPS: product states $\mathrm{Prod}(T)$, or MPS of bond dimension χ along the chain given by H's interaction graph in T, $\mathrm{MPS}_\chi(T)$. Covariance gives $\mathcal{S}(T_V) = V\mathcal{S}(T_0)$. Write $\mathcal{S} = \mathcal{S}(T_0)$ and let

$$E_{\mathcal{S}}(\phi) = 1 - \sup_{m\in\mathcal{S}} |\langle m|\phi\rangle|^2 = D(\phi, \mathcal{S})^2$$

be the geometric distance to the family; for $\mathcal{S} = \mathrm{Prod}$ this is the geometric measure of entanglement.

Definition 2 (Past-Hypothesis rule)**.** The *simplicity* of the division T_V is $\sigma(V) = \inf_t E_{\mathcal{S}(T_V)}(\psi(t)) = \inf_t E_{\mathcal{S}}(V^\dagger\psi(t))$. The rule selects the divisions of minimal simplicity index (or, with tolerance, all V with $\sigma(V) \le \sigma_{\min} + \epsilon$).

Lemma 3 (Invariance)**.** *The statement "H is K-local in T and some $\psi(t)$ lies within η of $\mathcal{S}(T)$" is a property of the triple (H, ψ, T) that is invariant under simultaneous global unitaries $(H, \psi, T) \mapsto (WHW^\dagger, W\psi, WTW^\dagger)$.*

Proof. Locality, time evolution and the family $\mathcal{S}(T)$ are all covariant under W, and overlaps are invariant. □

The rule is thus the same kind of object as the invariant properties of Ref. [10], but with a physical meaning: it is the Past Hypothesis. Since $\sigma(U(s)V) = \sigma(V)$, the selected set is always a union of time-translation orbits. The question is whether it is *only* the orbit of the true division.

Remark 4 (Time versus worlds). If the gaps $E_k - E_1$ are rationally independent, $\{U(t)\}$ is dense and equidistributed in Γ modulo phase (Kronecker, Weyl). Every T_V is then a limit of time translates, and averages over TPSs equal time averages along the history [6, Remark 11]. If instead $E_k = p_k\omega$

with integers p_k [15], the orbit is a closed loop of Haar measure zero in the $(N-1)$-dimensional torus. Almost every T_V then reads ψ as a state the history never visits. In a 3-qubit example with eigenvalues rounded to multiples of $\omega = 0.05$, random elements of Γ stayed at median distance 0.53 from the loop however long the history ran, whereas for the unrounded spectrum the distance kept shrinking. Section 3 shows that the Past Hypothesis removes this gap between the two cases.

3 Reduction to constrained tomography and exact selection

By Lemma 1, $\sigma(V)$ depends on V only through readings in the profile fibre of ψ, which is the same for every moment. Hence $\sigma_{\min} = \inf\{E_{\mathcal{S}}(\chi) : W(\chi) = W(\psi)\}$, and the infimum is attained because the fibre is compact and $E_{\mathcal{S}}$ is continuous. If the actual history has an exactly simple moment $\psi_0 = \psi(t_0) \in \mathcal{S}$, which we call the *bounce* (after the low-entropy turning point of cyclic cosmologies), then $\sigma_{\min} = 0$, and the divisions of zero simplicity index are those whose readings of some moment lie in $\mathcal{S}$ with the profile $W(\psi_0)$. The rule selects only the history's time translates for *every* exactly simple bounce if and only if the energy measurement *discriminates* $\mathcal{S}$: no two distinct states of $\mathcal{S}$ share an energy profile. This is a question of constrained quantum tomography [16, 17], with the eigenbasis of H as the measurement and simplicity as prior information.

Mereology	Tomography
readings of ψ over equally local TPSs	states with a given outcome distribution
Past Hypothesis (simple bounce)	prior: the state lies in $\mathcal{S}$
no rival world	the measurement discriminates $\mathcal{S}$
robust selection	stable recovery
local H	a structured, non-generic basis

Theorem 5 (Exact selection; after Kech–Wolf)**.** *Let $\mathcal{S}$ be a semi-algebraic set of pure states, regarded as rank-one density matrices, of real dimension d, and suppose $N - 1 > 2d$. For every eigenbasis outside a Haar-null set, and every non-degenerate spectrum, the following holds. If some moment $\psi_0 = \psi(t_0)$ lies in $\mathcal{S}$, then every division of zero simplicity index has the form T_V with $V = U(t - t_0)V_0$, where $V_0 \in \Gamma$ acts as a global phase on the energy support of ψ_0. In particular, if ψ_0 has full energy support, the rule selects exactly the time translates of T_0.*

Proof. Kech and Wolf show that k generic von Neumann measurements discriminate all states of a semi-algebraic set $\mathcal{S}$ whenever $k(N-1) > 2 \dim \mathcal{S}$ [17]. Take $k = 1$. Here "generic" means that the failing measurements form a set of zero Hausdorff measure in the full-dimensional semi-algebraic set of measurement schemes, which they establish by showing it has lower dimension [17, App. A]. For von Neumann measurements this set is the manifold of orthonormal bases, on which full-dimensional Hausdorff measure and Haar measure have the same null sets, since both have smooth positive densities in local charts. The failing eigenbases therefore form a Haar-null set. If $\sigma(V) = 0$ then $V^\dagger \psi(t) \in \mathcal{S}$ for some t, and it shares the profile of $\psi_0 \in \mathcal{S}$, so $V^\dagger \psi(t) = e^{i\alpha}\psi_0$. Hence $V^\dagger U(t - t_0)$ fixes ψ_0 up to phase, which for a diagonal unitary means it is a global phase on the support of ψ_0. □

Theorem 6 (Time-reversal-invariant case)**.** *Let $\mathcal{S}$ be a semi-algebraic set of pure states, regarded as rank-one density matrices, of real dimension d, and suppose $N - 1 > 2d$. For every real orthogonal eigenbasis outside a Haar-null subset of $O(N)$, and every non-degenerate spectrum, if $m, m' \in \mathcal{S}$ share an energy profile then $m' = e^{i\alpha}m$ or $m' = e^{i\alpha}\bar{m}$. If moreover $\mathcal{S}$ is closed under complex conjugation in the product basis, and some moment $\psi_0 = \psi(t_0) \in \mathcal{S}$ has full energy support, then*

every division of zero simplicity index reads the history either as a time translate of $\psi(t)$ or as a time translate of its time reversal $\overline{\psi(-t)}$. The proof is in Appendix A.

For product states $d = 2n$, so the condition is $2^n - 1 > 4n$, i.e. $n \geq 5$. For $\chi = 2$ MPS with open boundaries, parameter counting modulo gauge and normalization gives $d = 8n - 10$, so the condition is $n \geq 7$. Below these thresholds some simple states have twins (we find them numerically at 3–4 qubits). A generic exactly simple bounce is still recovered there, but the rule is fragile: at 3 qubits any perturbation of the bounce lets a rival win. Above the thresholds it becomes robust (Sec. 6).

4 Robust selection for random eigenbases

Real bounces are only approximately simple. The following separation property makes Theorem 5 quantitative.

Definition 7 (Profile separation)**.** The eigenbasis of H satisfies $\mathrm{Sep}(\mathcal{S}, r_0, h)$ if $\mathrm{Hel}(W(m), W(m'))^2 \geq h$ for all $m, m' \in \mathcal{S}$ with $D(m, m') \geq r_0$.

Lemma 8 (Separation implies robust selection)**.** *Assume* $\mathrm{Sep}(\mathcal{S}, r_0, h)$ *and* $\eta < \sqrt{h}/2$*. Let* $\psi_0 = \psi(t_0)$ *satisfy* $E_{\mathcal{S}}(\psi_0) \leq \eta^2$*. Then every reading* $\chi = V^\dagger \psi(t)$ *with* $E_{\mathcal{S}}(\chi) \leq \eta^2$ *— in particular every division the rule selects — satisfies* $D(\chi, \psi_0) < r_0 + 2\eta$*. Moreover* $D\big(V^\dagger \psi(t+s), \psi(t_0+s)\big) < r_0 + 2\eta$ *for all* s*: the selected division reproduces the whole history, up to a time shift, within* $r_0 + 2\eta$ *at every moment.*

Proof. Choose $m, m' \in \mathcal{S}$ with $D(m, \psi_0) \leq \eta$ and $D(m', \chi) \leq \eta$. For any states, $\mathrm{Hel}(W(\phi), W(\phi')) \leq D(\phi, \phi')$ (Lemma 16). Since $W(\chi) = W(\psi_0)$ by Lemma 1, the triangle inequality gives $\mathrm{Hel}(W(m), W(m')) \leq 2\eta < \sqrt{h}$, so $D(m, m') < r_0$. Then $D(\chi, \psi_0) \leq D(\chi, m') + D(m', m) + D(m, \psi_0) < r_0 + 2\eta$. For the second claim, $V^\dagger U(t - t_0) \in \Gamma$ commutes with $U(s)$, so $D(V^\dagger \psi(t+s), \psi(t_0+s)) = D(V^\dagger \psi(t), \psi_0)$. □

Theorem 9 (Product bounces, Haar eigenbasis)**.** *Let the eigenbasis of* H *be Haar-distributed, the spectrum arbitrary and non-degenerate, and* $0 < r_0 \leq 1$*. With probability at least*

$$1 - \Big(\frac{227\, n}{r_0^2}\Big)^{6n} \exp\Big(-\frac{2^n r_0^4}{19200}\Big),$$

$\mathrm{Sep}(\mathrm{Prod}, r_0, 0.114\, r_0^2)$ *holds. On this event the conclusions of Lemma 8 hold simultaneously for every state* ψ *and every* $\eta \leq 0.16\, r_0$*.*

Theorem 10 (MPS bounces, Haar eigenbasis)**.** *Under the same assumptions, with probability at least*

$$1 - \Big(\frac{566\, n}{r_0^2}\Big)^{8n\chi^2} \exp\Big(-\frac{2^n r_0^4}{19200}\Big),$$

$\mathrm{Sep}(\mathrm{MPS}_\chi, r_0, 0.114\, r_0^2)$ *holds, with the same consequences.*

Proofs are in Appendix A. Two ingredients carry them. First, an *exact* identity: for fixed states with overlap s and Haar U, $\mathbb{E} \sum_k |\langle k|m\rangle||\langle k|m'\rangle| = \frac{\pi}{4}\, {}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; |s|^2)$ at every N, which gives $\mathbb{E}\,\mathrm{Hel}^2 \geq (1 - \pi/4) D^2$ (Lemma 17). Second, concentration on the unitary group together with a net over the simple family, whose size grows only exponentially in $n\chi^2$ against concentration exponential in 2^n. The constants are loose: at $r_0 = 1$ the bound of Theorem 9 becomes non-trivial

at $n = 25$. This is intrinsic to the method rather than to our bookkeeping. Any union bound over a net of the simple family pays $\Theta(n \log n)$ in the exponent, against concentration of order $2^n r_0^4$, so even with optimistic constants (Gaussian-type concentration and minimal nets) the bound needs $n \gtrsim 19$. The theorems are therefore asymptotic statements, with failure probability $e^{-\Omega(2^n)}$; small sizes are covered by the simulations of Sec. 6. The spectrum enters nowhere, so the results hold for the commensurable spectra of Ref. [15]: the Past Hypothesis restores the "when, not which world" reading that density alone provides only for generic spectra (Remark 4).

Corollary 11 (Area-law bounces)**.** *Ground states of gapped one-dimensional local Hamiltonians are within trace distance η of MPS_χ with χ growing at most polynomially in n/η [20, 21]. For such bounces and a Haar eigenbasis, the conclusions of Lemma 8 therefore hold with probability $1 - e^{-\Omega(2^n)}$, although the bounce's entanglement grows with n.*

5 Local Hamiltonians

Eigenbases of local Hamiltonians form a Haar-null set, so Theorems 5–10 do not apply to them directly. In particular, those theorems are not evidence that local Hamiltonians satisfy Sep. For local H we establish only an obstruction (Proposition 12) and a conditional result (Proposition 14); everything else is numerical evidence (Section 6). The next result shows why a proof for local H must be of a different kind.

Proposition 12 (Coarse data cannot separate)**.** *Let $H = \sum_i H_i$ act on a lattice of n sites in fixed dimension, where each H_i satisfies $\|H_i\| \le 1$ and acts only on sites within a fixed distance of site i. Let q be a product state and $q' = Lq$, where L is a product of single-site unitaries on $O(1)$ sites, and suppose both energy variances are at least cn. Then the Kolmogorov distance between $W(q)$ and $W(q')$ (as distributions over energy) is $O(\mathrm{polylog}(n)/\sqrt{n})$, even if $\langle q|q'\rangle = 0$.*

Proof. The Berry–Esseen theorem for quantum lattice systems [27, 28, 26], whose hypotheses are those stated, says that the energy distribution of a state with finite correlation length, in particular a product state, is within $O(\mathrm{polylog}(n)/\sqrt{n})$ in Kolmogorov distance of the Gaussian with the same mean and variance. Changing $O(1)$ sites changes $O(1)$ local terms, so the means and variances differ by $O(1)$, while the standard deviations are of order $\sqrt{n}$. Two such Gaussians are $O(1/\sqrt{n})$ apart in Kolmogorov distance; apply the triangle inequality. □

Numerically, the exact profiles of such pairs stay far apart ($\mathrm{Hel}^2 \approx 0.15$–$0.2$ at 8–11 qubits), so the separating information lies in eigenvector structure below the resolution of central-limit methods. Equivalently, it lies in the late-time fluctuations of the return amplitude $\langle q|e^{-iHt}|q\rangle$, beyond Lieb–Robinson and cluster-expansion techniques. Some input about eigenvector fine structure, of eigenstate-thermalization type [29, 30], is therefore unavoidable.

Hypothesis 13 (ETH-A: random amplitudes in energy shells)**.** For m, m' in the simple family, the amplitudes $a_k = \langle k|m\rangle$ and $b_k = \langle k|m'\rangle$ are modelled as independent across k, with (a_k, b_k) circular complex Gaussian with $\mathbb{E}|a_k|^2 = \rho_k$, $\mathbb{E}|b_k|^2 = \rho'_k$ and $\mathbb{E}\bar{a}_k b_k = c_k\sqrt{\rho_k \rho'_k}$, where ρ, ρ' and c vary smoothly with energy.

Proposition 14 (Separation under ETH-A)**.** *Under Hypothesis 13, let $\mathrm{BC} = \sum_k \sqrt{\rho_k \rho'_k}$ and let $\gamma = \mathrm{BC}^{-1} \sum_k \sqrt{\rho_k \rho'_k}\,(1 - |c_k|^2)$. Then*

$$1 - \mathbb{E}\sum_k |a_k||b_k| \;\ge\; (1 - \mathrm{BC}) + \left(1 - \tfrac{\pi}{4}\right)\mathrm{BC}\,\gamma.$$

Hence, if $\gamma \ge \gamma_0 D(m, m')^2$ for all simple pairs, then $\mathbb{E}\,\mathrm{Hel}^2 \ge (1 - \pi/4)\gamma_0 D^2$.

The proposition bounds the *expected* separation of each pair; it is not a uniform statement over the simple family.

Remark 15 (Uniform version: a sketch, not a result). If the Gaussian model were extended jointly to the whole simple family, Gaussian concentration and the nets of Appendix A would give $\mathrm{Sep}(\mathcal{S}, r_0, (1-\pi/4)\gamma_0 r_0^2 - \epsilon)$ with failure probability $\exp\big(O(n\chi^2 \log n) - \Omega(N_{\mathrm{eff}}\,\epsilon^2)\big)$, where $N_{\mathrm{eff}} = 1/\max_k \rho_k$. Hypothesis 13 specifies only pairwise statistics, so this requires an additional modelling assumption, which we do not formalize.

Proof. For each k, $\mathbb{E}|a_k||b_k| = \sqrt{\rho_k \rho'_k}\,\frac{\pi}{4} F(|c_k|^2)$ with $F = {}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; \cdot)$ (Lemma 17). F is convex with $F(0) = 1$ and $F(1) = 4/\pi$, so $F(x) \le \frac{4}{\pi} - (\frac{4}{\pi} - 1)(1 - x)$. Summing gives the display. Since $x := (1 - \pi/4)\gamma \le 1$, we have $(1 - \mathrm{BC}) + \mathrm{BC}\, x \ge x$. □

If in addition $\rho_k = \rho'_k = 1/N$ and c_k does not depend on k, as for Haar bases, then $c_k = \langle m|m'\rangle$, because $\sum_k \bar{a}_k b_k = \langle m|m'\rangle$ holds identically. Then $\gamma = D^2$ and the Haar bound is recovered. ETH-A alone does not force c_k to be constant. At finite temperature, ETH suggests that γ_0 is controlled by thermal fluctuations of local operators and stays positive; Section 6 measures it.

6 Numerical evidence

We use generic nearest-neighbour 2-local chains with Gaussian couplings on all Pauli pairs and fields (Appendix C). Commensurable versions round each eigenvalue to a multiple of $\omega = 0.05$, which changes H by at most 0.025 in operator norm and leaves the eigenvectors unchanged. The rule is optimized over Γ by multi-start BFGS. A pick counts as *on-history* if it lies in the basin of the true bounce or has fidelity ≥ 0.99 with some $\psi(t)$. The product criterion is the mean single-qubit entropy; the MPS criterion is the discarded Schmidt weight at $\chi = 2$.

Thresholds. We test two kinds of bounce (Table 1): *perturbed product* states $\psi_0 = \cos\theta\, p + \sin\theta\, r$, with p a random product state and r a random state, and *ground states* of $-\sum ZZ - g\sum X - 0.1\sum Z$, which have realistic short-range entanglement. The product rule fails at 3 qubits, where exact twins exist, and the tolerated entanglement then rises steadily with size; by 7 qubits every ground-state bounce tested is recovered. The MPS rule shows the same transition shifted to 6–7 qubits, as the dimension count predicts. Searched separation constants $\kappa = \inf \mathrm{Hel}/D$ over product pairs are zero at 3–4 qubits, where nearly orthogonal pairs share profiles, and positive from 5 qubits on. For local H the only near-degenerate directions found were short time translations.

ETH-A and rival floors. For product pairs $q' = Lq$, with L a rotation on 1–4 sites, the ETH-A model (envelopes and correlations smoothed at energy resolution 0.5) reproduces exact Hel^2 at 8–11 qubits with mean absolute error 0.004–0.009, about 3% of Hel^2, against 0.008–0.021 for the Haar formula (Fig. 1). The fitted ratio γ/D^2 has median about 0.92 at mid-spectrum and minima 0.44–0.49 for the coldest product states of three 10-qubit chains, so γ_0 falls at low temperature but stays away from zero. Separately, the lowest per-site entropy reached by any off-history reading of an unentangled bounce rises with size for both local and Haar eigenbases, from 0.04–0.08 bits at 4 qubits to 0.23–0.44 at 7.

Search adequacy. Extra random starts (about 2,700 per 5-qubit model, 165–290 per 7-qubit case) found nothing better than the true bounce. The weakest points are 8-qubit perturbed bounces at $\theta = 0.8$, where random starts rarely reached even the true basin, so rivals may have been missed.

qubits	product rule		MPS rule ($\chi = 2$)	
	perturbed: tolerated entanglement	ground states	ground states	perturbed ground states
3	0.00 (0.00–0.01)	12/36	–	–
4	0.27 (0.06–0.48)	14/36	–	–
5	0.32 (0.18–0.74)	29/36	6/9	0/12
6	0.57 (0.18–0.59)	30/36	9/9	7/12
7	0.77 (0.67–0.78)	36/36	12/12	11/12
8	none up to 0.82	6/6*	6/6	–

Table 1: Past-Hypothesis rule. Perturbed product bounces: median (range) over six models of the bounce entanglement (mean single-qubit entropy, bits) at which a rival first wins; two models at 8 qubits. Other columns: cases in which the rule picks the history. Product-rule ground states use $g \in \{3, 2, 1.5, 1.2, 1, 0.8\}$ (bounce entanglement 0.08–0.32 bits), MPS ground states $g \in \{1.5, 1.2, 1.0\}$, and MPS perturbations $\theta = 0.1$–0.45. *The three hardest fields only.

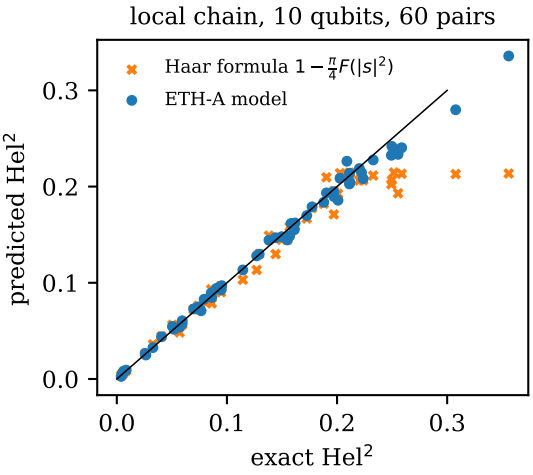


Figure 1: Exact versus predicted Hel^2 for 60 product pairs on a 10-qubit chain (mean absolute error 0.005 for ETH-A, 0.013 for the Haar formula).

7 Local Hamiltonians: mechanism and worst cases

A route through random-matrix theory. For $H = H_{\mathrm{loc}} + \lambda W$ with W a Wigner matrix, ETH and joint Gaussian fluctuations of eigenvector overlaps are theorems [22, 23, 24, 25], giving exactly the structure ETH-A assumes. Turning this into a proof of Sep would need versions uniform over all N eigenvectors with failure probability $e^{-\Omega(n \log n)}$ per pair, and for deformations as large as H_{loc}, which the published results do not provide.

A short-time mechanism. The correlations $c(E)$ are built from energy-resolved matrix elements whose Fourier transforms involve $\langle q|e^{-iHt}|q'\rangle$. For product states these amplitudes decay on the time scale $1/\sqrt{n}$, so at any resolution that is not exponentially small the correlations are set by short-time dynamics. For states differing on a few sites, a cluster expansion then suggests $c(E) \approx \langle q|q'\rangle$, hence $\gamma \approx D^2$: the random-matrix value. This argument is heuristic. Consistent with it, for random product states and partners differing on one or two sites, the mean separation matches the random-matrix prediction $1 - \frac{\pi}{4} F(1 - D^2)$ to within 8% at 7–11 qubits, with or without added GUE noise ($\lambda \leq 2$).

Worst cases and energy dependence. Random pairs can miss the hardest ones, so we also minimized the separation ratio adversarially over partners differing on the two middle sites, with $D^2 \geq 0.1$ (Table 2). For chaotic chains near mid-spectrum the worst case rises steadily, reaching 0.91 of the random-matrix value at 12 qubits. An integrable control, a random-coupling transverse-field Ising chain, is consistently lower but does not collapse. Separation weakens at low energy but improves with size at every energy tested. Much of the energy dependence is explained by the effective dimension $N_{\mathrm{eff}} = 1/\sum_k p_k^2$ of the base state (Spearman correlation 0.62): at 10% of the spectrum, product states at these sizes spread over only 4–8 eigenstates, against 25–430 near mid-spectrum. At fixed N_{eff}, larger systems still separate better, and rare weak couplings do not explain the hard cases (correlation 0.07). The conjecture that chaotic local systems attain the random-matrix value is thus well supported at high energy and open at low energy. When the partner may differ on every site, the worst cases are lower. Searches from mid-spectrum and low-energy base states give 0.04–0.21 at 6 qubits, 0.24–0.30 at 8 and 0.23–0.26 at 10, with strongly entangling relating unitaries. These stay well away from zero but show no clear trend from 8 to 10 qubits, so for general partners only the weaker conjecture, a size-independent $c > 0$, is supported.

	chaotic, energy fraction			integrable
qubits	10%	25%	mid-spectrum	mid-spectrum
6	0.36 (0.13)	0.43 (0.21)	0.46 (0.15)	0.28 (0.18)
8	0.57 (0.31)	0.61 (0.47)	0.65 (0.48)	0.47 (0.16)
10	0.60 (0.33)	0.73 (0.46)	0.78 (0.73)	0.56 (0.35)
12	–	–	0.91 (0.87)†	–

Table 2: Adversarial worst-case separation as a fraction of the random-matrix value: median (minimum) over 12–16 base states from 3–4 Hamiltonians per entry. Mid-spectrum columns use random product states. †One Hamiltonian, 4 base states, and a lighter search, which raises values by at most 0.04 at 10 qubits. Search-based values are upper bounds on the true worst case.

Symmetries. If H commutes with a product unitary P, such as a global spin flip, then q and Pq share a profile and can be far apart, so no separation constant exists. Such twins are harmless,

because $PT_0P^\dagger = T_0$; distances should be taken modulo them. Why the exact problem resists the standard tools is discussed in the extended version.

Time reversal. Antiunitary symmetries behave differently. If H is real in a product basis, as for time-reversal-invariant Hamiltonians without magnetic fields, then every state q and its complex conjugate $\bar{q}$ have identical profiles, and $\bar{q}$ is a product state whenever q is. The unitary in Γ relating them is in general entangling (operator entanglement 0.82–0.97 in our examples), so it changes the division. In that division the history reads as $\overline{\psi(-t)}$: the same history run backward. A search with partners free on all sites finds exactly these twins in real-coupled chains, with separation ratio below 10^{-9} at 6 and 8 qubits and each twin within 4×10^{-5} of a time-shifted $\bar{q}$; making the couplings of a complex chain real creates them. For time-reversal-invariant Hamiltonians the residual freedom is therefore time translation together with time reversal: a single simple moment fixes the history but not its direction. Theorems 5–10 are unaffected, since real eigenbases form a Haar-null set, and our main simulations use complex couplings. Theorem 6 makes this exact for generic real eigenbases. Numerically, random real bases have twins other than conjugates at 3–4 qubits and none from 5 qubits on: the minimum of Hel^2/D^2, with distance taken modulo conjugation, is 0.002–0.003 at 5 qubits and 0.013–0.047 at 7, matching complex bases. In real-coupled chaotic chains, an all-sites search excluding time shifts and time reversal finds worst ratios of 0.06–0.13 at 6–8 qubits, nonzero but lower than for complex chains.

8 Discussion

Relation to the Entanglement Past Hypothesis. Al-Khalili and Chen [3] propose that the early universe had low entanglement relative to some factorization. They regard the dependence on a factorization as a difficulty and note an apparent circularity: specifying the factorization by spatial structure conflicts with deriving space from entanglement. Our rule runs the other way. It takes the divisions in which H is equally local, fixed by (H, ψ) alone, and selects the one in which some moment is simple. By Theorem 5, for generic eigenbases and $n \geq 5$, a bounce with full energy support is unentangled in only one member of this class. Requiring low entanglement in *every* member therefore fails, while requiring it in *some* member selects the division. Locality is what gives the rule content: without it, a low-entropy past is trivial, since a generic state has some split in which its entropy starts low and grows [4].

Direction of time. For time-reversal-invariant Hamiltonians the rule fixes the history and its division into subsystems, but not the direction of time: the simple moment looks the same from either side. This fits the picture of Barbour, Koslowski and Mercati, in which solutions of a time-symmetric gravitational model divide at a unique "Janus point", with arrows of time pointing away from it in both directions [1], and the two-arrow cosmology of Carroll and Chen [2]. On this reading the simple moment determines what the parts are, while the direction of time is set on each side by entropy growing away from it. We have not found this connection made elsewhere, but have not searched exhaustively.

Objections. (i) *Simplicity smuggles in structure.* The rule is an invariant property of (H, ψ, T) (Lemma 3), close to the Entanglement Past Hypothesis. It is, however, a stronger and more specific postulate than low thermodynamic entropy. A critic can fairly say that we have not derived subsystem structure from the Past Hypothesis, but introduced a simplicity postulate and shown that it identifies a factorization. We accept that description. The content of the result is that

this postulate, unlike simplicity without the locality constraint, is unambiguous. (ii) *Stoica showed the Past Hypothesis cannot give uniqueness* [6, Sec. VII.B]. We agree: infinitely many divisions remain, but they are exactly the history's time translates. Stoica notes that if only time translation remained, a unique spacetime could be recovered in place of a unique space (Remark 10), and raises objections for dense time orbits (Remark 11). In Sec. VII.E he observes that restricting to a single orbit $\{e^{-iHt}\psi_0\}$ gives a unique spacetime, but calls the restriction ad hoc and asks for a derivation from physical principles. Our result can be read as such a derivation, and the selected divisions come ordered by t. We do not address his later argument that HSF cannot account for change [8], and the rule does not select a "now". (iii) *Random eigenbases are unphysical.* Proposition 12 shows that some eigenvector-level hypothesis is unavoidable for local H. ETH-A is a random-amplitude version of ETH and matches our data at 8–11 qubits. (iv) *The simulations are small.* For random eigenbases the large-n claims rest on the theorems; for local Hamiltonians they remain conjectural.

Scope. We work in finite-dimensional HSF with a non-degenerate spectrum and do not defend the framework. Degenerate and continuous spectra, field theory and gravity are not treated.

Open problems. (1) Prove Sep, modulo time translations, symmetries and, for time-reversal-invariant H, time reversal, for chaotic local Hamiltonians under assumptions weaker than ETH-A. Even the exact version, no twins for almost every choice of local couplings, is open. (2) Explain why the tolerated per-site entanglement and the rival floor grow with n. A heuristic in the Haar model suggests that no rival exceeds a per-site purity excess of order $\log n/n$, but correlations between single-site marginals must be controlled. (3) Determine γ_0 at low energy, where Table 2 shows the weakest separation.

Disclosure of AI use

This manuscript was produced by Claude (Anthropic) in an extended interactive session. The AI proposed the main constructions, proved the stated results, wrote and ran all numerical code, performed the verification audit of Appendix B, and wrote the text. The human user directed the project, asked for checks and corrections, and chose to post it for feedback. No human has yet taken responsibility for the correctness of the content; it should be read as an unreviewed AI-generated draft.

Code and data availability

All scripts and raw results are provided with this manuscript: `past_hypothesis_scan.py`, `mps_rule_scan.py`, `separation_constant.py`, `eth_check.py`, `random_matrix_prediction.py`, `mereology_toy_model.py` and `past_hypothesis_raw_results.tar.gz`.
An extended version of this paper, with all tables, both figures and the full verification record, accompanies the code bundle.

A Proofs

Lemma 16. *For any states,* $\mathrm{Hel}(W(\phi), W(\phi')) \leq D(\phi, \phi')$.

Proof. $\sum_k |\langle k|\phi\rangle||\langle k|\phi'\rangle| \geq |\langle\phi|\phi'\rangle|$, so $\mathrm{Hel}^2 \leq 1 - |\langle\phi|\phi'\rangle| \leq 1 - |\langle\phi|\phi'\rangle|^2$. □

Lemma 17 (Exact expectation)**.** *Let m, m' be unit vectors with $|\langle m|m'\rangle| = s$, and let $\{|u_k\rangle\}$ be the columns of a Haar-random $U \in U(N)$. Then, at every N,*

$$\mathbb{E}\sum_{k=1}^{N} |\langle u_k|m\rangle||\langle u_k|m'\rangle| = \frac{\pi}{4}F(s^2), \qquad F = {}_2F_1(-\tfrac{1}{2}, -\tfrac{1}{2}; 1; \cdot),$$

and consequently $1 - \mathbb{E}\sum_k |\langle u_k|m\rangle||\langle u_k|m'\rangle| \geq (1-\pi/4)(1-s^2)$.

Proof. Write $m' = e^{i\alpha}(s\, m + c\, m_\perp)$ with $c = \sqrt{1-s^2}$ and $m_\perp \perp m$. Each column u_k is uniform on the sphere, so $(\langle u_k|m\rangle, \langle u_k|m_\perp\rangle)$ has the law of $(\bar{g}_1, \bar{g}_2)/\|g\|$ for a standard complex Gaussian vector $g \in \mathbb{C}^N$ with $\mathbb{E}|g_i|^2 = 1$. The direction $g/\|g\|$ is independent of $\|g\|$, and $\mathbb{E}\|g\|^2 = N$. The integrand is homogeneous of degree 2, so $\mathbb{E}[|x_1||sx_1 + cx_2|] = N^{-1}\mathbb{E}|g_1||sg_1 + cg_2|$. It remains to show $\mathbb{E}|X||Y| = \frac{\pi}{4}F(s^2)$ for $X = g_1$, $Y = sg_1 + cg_2$. Conditionally on X, Y is complex Gaussian with mean sX and variance c^2, so $|Y|$ is Rice-distributed with $\mathbb{E}[|Y|\,|\,X] = c\frac{\sqrt{\pi}}{2}\,{}_1F_1(-\frac{1}{2}; 1; -s^2|X|^2/c^2)$. With $u = |X|^2 \sim \mathrm{Exp}(1)$ and the Laplace transform $\int_0^\infty u^{b-1}e^{-u}\,{}_1F_1(a; 1; -ku)\,du = \Gamma(b)\,{}_2F_1(a, b; 1; -k)$, $\mathbb{E}|X||Y| = c\frac{\pi}{4}\,{}_2F_1(-\frac{1}{2}, \frac{3}{2}; 1; -s^2/c^2)$. Pfaff's transformation ${}_2F_1(a, b; c; z) = (1-z)^{-a}\,{}_2F_1(a, c-b; c; \frac{z}{z-1})$ with $z = -s^2/c^2$ gives $(1-z)^{1/2} = 1/c$ and $z/(z-1) = s^2$, hence $\mathbb{E}|X||Y| = \frac{\pi}{4}F(s^2)$; the case $s = 1$ follows by continuity. Summing over k gives the first claim. For the second, F has positive Taylor coefficients, so it is convex, and Gauss's theorem gives $F(1) = 4/\pi$. The function $G(x) = \frac{\pi}{4}(1-F(x)) + (1-\frac{\pi}{4})x$ is therefore concave with $G(0) = G(1) = 0$, hence $G \geq 0$ on $[0, 1]$. □

Lemma 18 (Lipschitz bounds and concentration)**.** *Let $g_{m,m'}(U) = \sum_k |\langle u_k|m\rangle||\langle u_k|m'\rangle|$. Then $|g(U) - g(U')| \leq 2\|U - U'\|_{\mathrm{HS}}$, hence $\Pr[g \geq \mathbb{E}g + t] \leq e^{-Nt^2/48}$. Moreover $|g_{m,m'} - g_{\tilde{m},\tilde{m}'}| \leq \sqrt{2}\left(D(m, \tilde{m}) + D(m', \tilde{m}')\right)$ for every U.*

Proof. By the triangle and Cauchy–Schwarz inequalities, $|g(U) - g(U')| \leq \|(U-U')^\dagger m\| + \|(U - U')^\dagger m'\| \leq 2\|U - U'\|_{\mathrm{op}} \leq 2\|U - U'\|_{\mathrm{HS}}$. For Haar-distributed $U \in U(N)$ with the Hilbert–Schmidt metric, every L-Lipschitz function satisfies $\Pr[f \geq \mathbb{E}f + t] \leq e^{-Nt^2/(12L^2)}$ [31, Cor. 17]; see also [35, Ch. 5]. For the last claim, $|g_{m,m'} - g_{\tilde{m},m'}| \leq \min_\varphi \|m - e^{i\varphi}\tilde{m}\| = \sqrt{2 - 2|\langle m|\tilde{m}\rangle|} \leq \sqrt{2}\, D(m, \tilde{m})$. □

Lemma 19 (Nets)**.** *For $0 < \delta \leq 1$, Prod has a δ-net in D of size at most $(2n/\delta)^{3n}$, and MPS_χ has one of size at most $(5n/\delta)^{4n\chi^2}$.*

Proof. For product states, $D(\otimes_i q_i, \otimes_i \tilde{q}_i) \leq \sum_i D(q_i, \tilde{q}_i)$ by subadditivity of the trace distance under tensor products, and $D = |r - \tilde{r}|/2$ for Bloch vectors. A $(2\delta/n)$-net of the unit sphere in $\mathbb{R}^3$ has at most $(1 + n/\delta)^3 \leq (2n/\delta)^3$ points; take products. For MPS in left-canonical form, ψ is a product of isometries V_i, each with real dimension at most $4\chi^2$. Take as net a maximal ϵ'-separated subset of the isometries themselves, which is an ϵ'-net in operator norm consisting of isometries. Balls of radius $\epsilon'/2$ about its points are disjoint and lie in the operator-norm ball of radius $1 + \epsilon'/2$, so it has at most $(1 + 2/\epsilon')^{4\chi^2} \leq (5/\epsilon')^{4\chi^2}$ points. Because the net points are isometries, every net state $\tilde{\psi}$ is a normalized MPS in left-canonical form, and every factor in both chains has operator norm 1. Telescoping then gives $\|\psi - \tilde{\psi}\| \leq \sum_i \|V_i - \tilde{V}_i\|_{\mathrm{op}}$, and $D \leq \|\psi - \tilde{\psi}\|$ for unit vectors. Take $\epsilon' = \delta/n$. □

Proof of Theorems 9 and 10. Set $\epsilon = 0.1r_0^2$, $t = \epsilon/2$ and $\delta = \epsilon/(8\sqrt{2})$, and let $\mathcal{N}$ be the net of Lemma 19. By Lemma 18 and a union bound, with probability at least $1 - |\mathcal{N}|^2 e^{-Nt^2/48}$ every net pair satisfies $g \leq \mathbb{E}g + t$. Every pair lies within δ of a net pair in each argument, and g and $\mathbb{E}g$ each move by at most $2\sqrt{2}\,\delta$. So on this event $g \leq \mathbb{E}g + t + 4\sqrt{2}\,\delta = \mathbb{E}g + \epsilon$ for all pairs. By Lemma 17, every pair with $D \geq r_0$ then has $\mathrm{Hel}^2 = 1 - g \geq (1-\pi/4)r_0^2 - 0.1r_0^2 \geq 0.114\, r_0^2$. Inserting $|\mathcal{N}|^2 \leq (16\sqrt{2}\, n/\epsilon)^{6n}$ for products, or $(40\sqrt{2}\, n/\epsilon)^{8n\chi^2}$ for MPS, and $Nt^2/48 = 2^n r_0^4/19200$, gives the stated probabilities. Finally, $\eta \leq 0.16 r_0$ implies $2\eta < \sqrt{0.114}\, r_0$, so Lemma 8 applies. □

Proof of Theorem 6. For a real unit vector o and $m = a + ib$ with a, b real, $|\langle o|m\rangle|^2 = o^{\mathsf{T}}(aa^{\mathsf{T}} + bb^{\mathsf{T}})o = o^{\mathsf{T}}\,\mathrm{Re}(mm^\dagger)\,o$. So m and m' share a profile in the real basis $O = (o_1, \ldots, o_N)$ if and only if $\Phi(O) := \mathrm{diag}(O^{\mathsf{T}}AO) = 0$, where $A = \mathrm{Re}(mm^\dagger) - \mathrm{Re}(m'm'^\dagger)$ is real symmetric and traceless.

$A = 0$ exactly for the stated twins. $A = 0$ means $aa^{\mathsf{T}} + bb^{\mathsf{T}} = a'a'^{\mathsf{T}} + b'b'^{\mathsf{T}}$, so $(a', b') = (a, b)R$ for some $R \in O(2)$. Rotations give $m' = e^{i\alpha}m$ and reflections give $m' = e^{i\alpha}\bar{m}$; if a, b are dependent, m is real up to phase and the two cases coincide.

For $A \neq 0$ the fibre $\Phi^{-1}(0)$ has codimension at least $N - 1$ in $O(N)$. With $B = O^{\mathsf{T}}AO$ and X skew-symmetric, $d\Phi_O(OX) = \mathrm{diag}([B, X])$, and each pair $j < k$ contributes $2B_{jk}X_{jk}(e_k - e_j)$. Hence $\mathrm{rank}\, d\Phi_O = N - c(B)$, where $c(B)$ is the number of connected components of the graph with edges $\{j, k\}$ for $B_{jk} \neq 0$. Where the graph is connected, Φ is a submersion onto the hyperplane $\sum_k x_k = 0$ and the fibre has codimension $N - 1$. Where it is disconnected, the columns indexed by a component I (complement J) span an A-invariant subspace $W \oplus K$, with W in the range of A ($\dim W = w$, $\mathrm{rank}\, A = r \geq 1$) and K in its kernel. The conditions $B_{IJ} = 0$ then have codimension $w|J| + (r - w)|I| - w(r - w)$, and the diagonal conditions not already implied add at least $N - 2$, or $|J| - 1$ when $w = 0$ (and $|I| - 1$ when $w = r$). In every case the total is at least $N - 1$. This parameter count assumes the remaining diagonal conditions are independent within the stratum; we have checked this step less carefully than the others, and it is the first a referee should examine.

Conclusion. The incidence set $\{(O, m, m') : A \neq 0,\ \Phi(O) = 0\}$ is semi-algebraic, of dimension at most $\dim O(N) - (N-1) + 2d < \dim O(N)$. Its projection to $O(N)$ therefore has lower dimension and is Haar-null, and outside it profile equality in $\mathcal{S}$ forces $A = 0$. For the statement about divisions, argue as in Theorem 5. If $V^\dagger\psi(t) = e^{i\alpha}\bar{\psi}_0$, then, because H is real, $V^\dagger\psi(t + s) = e^{i\alpha}e^{-iHs}\bar{\psi}_0 = e^{i\alpha}\,\overline{\psi(t_0 - s)}$. □

B Verification record

Every computable claim was re-checked independently of its derivation. The checks were adversarial searches for counterexamples to each inequality, 30-digit arithmetic for the key identity and each step of its derivation, exact arithmetic for every constant, Jacobian-rank checks of the dimension counts, a level-statistics check that the test Hamiltonians are chaotic ($\langle r\rangle = 0.587$ and 0.599, against the GUE value 0.600), and recomputation of every table from raw data or deterministic reruns. This found nine errors in reporting and wording, all corrected, none affecting a theorem or conclusion.

The mathematics (Sections 2–5 and Appendix A) was then reviewed over three rounds by AI models from other developers. They verified Lemmas 17, 18 and 8, Theorems 5 and 9, and the expectation bound of Proposition 14. They identified four gaps, all repaired: the MPS net, the hypotheses of Proposition 12, the state space in Theorem 5, and an infinite-temperature claim. The third round confirmed the repairs. An AI review is a screen, not an expert review.

Every reference was checked to exist and to say what it is cited for, including the cited passages of Ref. [6]. Not verified: the proofs of cited theorems, novelty, and the proofs as a whole by a human expert. The full record, with scripts and logs, is in the extended version and the code bundle.

Counterexample search. A later search with partners free on all sites found exact twins in time-reversal-invariant chains. They are time-reversed readings of the same history, discussed in Sec. 7. It found no twins in complex-coupled chaotic chains, where the worst cases stay at 0.23–0.30 of the random-matrix value at 8–10 qubits. Theorem 6 and its proof were added after the external reviews and have not been reviewed; the threshold it predicts was checked numerically.

C Numerical methods

The Hamiltonian is $H = \sum_i \sum_a h_{i,a}\sigma_i^a + 0.7\sum_i \sum_{a,b} J_{i,ab}\sigma_i^a\sigma_{i+1}^b$, with independent standard normal couplings and a seed per model. Seeds are small consecutive integers recorded in the code, so every table can be regenerated and extended to other seeds. Eigenvectors come from exact diagonalization. For the product rule, the smooth objective is the mean single-qubit linear entropy of $V^\dagger\psi_0$, and results are reported in von Neumann entropy. For the MPS rule, the objective is $\sum_{\text{cuts}}(1 - \sum_{j\le\chi} s_j^2)$, whose gradient is $-2\,\mathrm{Re}$ of the overlap with the truncated state at each cut. Both objectives are optimized over $\varphi \in [0, 2\pi)^N$ with BFGS from the identity (the true basin) and from 6–2,700 random starts, depending on size. A threshold is the midpoint between the last bounce entanglement at which every tested point picked the history and the first at which one did not. Separation constants minimize Hel/D (or Hel/D_T, with D_T minimized over $|t| \le 0.6$) over pairs of product states by L-BFGS from random starts. The ETH-A model uses envelopes and correlations smoothed with a Gaussian kernel of width 0.5 in energy.

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